

Introduction to python-TRM

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In this brief tutorial, we introduce python-TRM. This package offers tools to explore evolutionary dynamics under the Territorial Raider Model. This framework allows the study of structured multiplayer interactions under social dilemmas. See [Pires and Broom, 2026] for a description of the model and its use. The foundational basis of this framework can be found in [Broom and Rychtář, 2012] and [Broom et al., 2015], and its more recent developments and uses in [Pattni et al., 2017], [Pires and Broom, 2024] and [Pires, 2024].

Example Usage of TRM

In this section we provide a fully documented example illustrating how to use the TRM package to construct a population under the Territorial Raider Model (TRM), compute fixation probabilities, and generate fixation curves across different parameter values.

Importing required modules

```
import TRM
import numpy as np
```

The first line imports the TRM package, giving access to all population- and fixation-related methods. The numpy library is imported to construct the adjacency matrix representing the spatial structure of the population.

Population and model parameters

The following block lists the key parameters used by TRM:

- M : number of sites (locations) in the spatial network.
- h : home fidelity parameter.
- w : intensity of selection.
- C, V : payoffs (cost and value).
- `adjacency_matrix`: connectivity among sites.

- game: evolutionary game used (e.g., PD, CPD, SD, HD, etc.).
- dynamics: update mechanism (BDB, DBB, LB, etc.).
- Q : subpopulation size per site.
- ω : parameter for PDV-type games.
- L : threshold for Stag-Hunt/FSH/TVD/TSD games.
- topology: name of spatial network.

These parameters define both the strategic and spatial behaviour of individuals.

Building the adjacency matrix

```
M=5
adjacency_matrix = np.zeros((M, M), dtype=int)
for i in range(M):
    adjacency_matrix[i, (i-1) % M] = 1
    adjacency_matrix[i, (i+1) % M] = 1
```

We create a ring (cycle) network of $M = 5$ locations. The adjacency matrix is initialized as a square $M \times M$ matrix of zeros. Each node is connected to its immediate left and right neighbours, generating the structure.

Constructing the population

```
pop = TRM.Population(
    M=M, h=10., w=0.3, C=1., V=5.,
    game="CPD", dynamics="DBB",
    adjacency_matrix=adjacency_matrix, Q=2,
    topology="circle")
```

A Population object is instantiated with:

- $M = 5$ spatial sites,
- home fidelity $h = 10$,
- selection intensity $w = 0.3$,
- payoff parameters $C = 1, V = 5$,
- the CPD (Charitable Prisoner's Dilemma) game,
- DBB evolutionary dynamics (death–birth process with selection acting on death),
- adjacency matrix defined above,
- subpopulation size $Q = 2$ at each site.

This fully defines the behavioural, evolutionary, and spatial rules of the system.

Computing fixation probabilities

```
fix_prob = pop.get_fixation_probabilities()
```

The method `get_fixation_probabilities()` returns a 2x2 matrix:

$$\begin{bmatrix} \rho_{C \rightarrow D} & \rho_{D \rightarrow C} \\ \rho_{C \rightarrow D}^T & \rho_{D \rightarrow C}^T \end{bmatrix},$$

where the top row contains arithmetic fixation probabilities and the bottom row contains temperature-weighted fixation probabilities.

Generating fixation curves across home fidelity values

```
TRM.get_fixation_h(M=M, w=0.4, C=1., V=1.,
    game="CPD", dynamics="DBB",
    adjacency_matrix=adjacency_matrix, Q=2,
    topology="circle")
```

The function `get_fixation_h` computes fixation probabilities for a range of home fidelity values $h \in [0.01, 100]$ on a logarithmic scale. The results are automatically saved into a data/ directory. The example repeats this computation for multiple values of V (1, 2, 5, 10), allowing comparison of different payoff regimes.

```
TRM.get_fixation_h(... V=2 ...)
TRM.get_fixation_h(... V=5 ...)
TRM.get_fixation_h(... V=10 ...)
```

This produces a full dataset capturing how h and V jointly shape the fixation behaviour of cooperators and defectors.

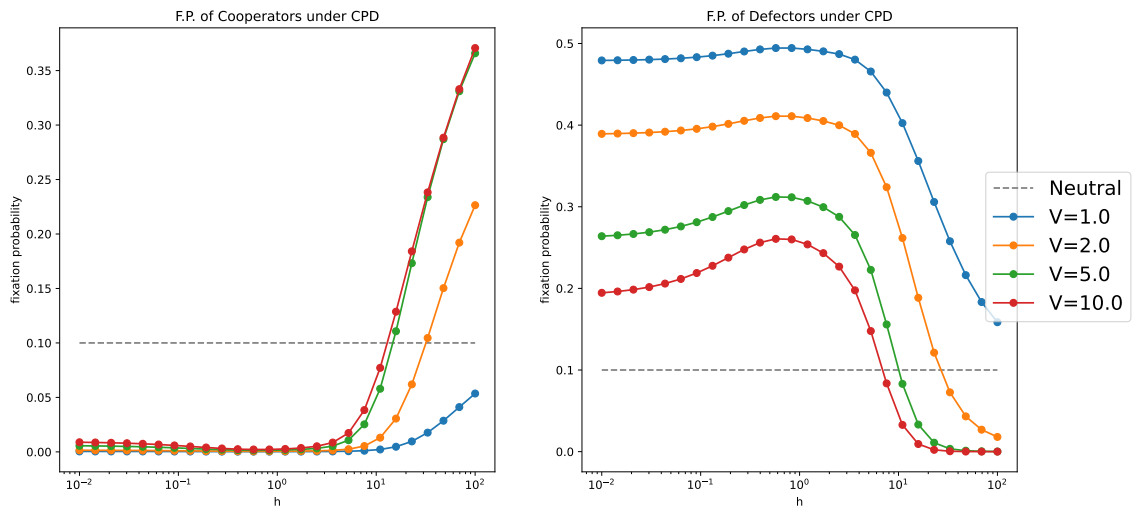


Figure 1: Fixation probability as a function of home fidelity under the Territorial Raider Model.

The package repository is available at <https://github.com/diogolpires/TRM>, which contains an example folder. This folder includes the full example code in `run_fixation.py` and the generated data in `data/`. Figure based on this data can be produced using `plot_fixation.py`, which saves the plot to `plots/`. The example plot generated is shown in figure 1.

References

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